STAT 578: Spectral Methods, Network Analysis, and Nonconvex Optimization

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Lecture 5
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1 Davis–Kahan Theorem

1.1 Setup

Let $M \in \mathbb{R}^{n\times n}$ be symmetric with spectral decomposition

$$M = \sum_{i=1}^{r} \lambda_i u_i u_i^\top, \qquad U = [u_1, \dots, u_r] \in \mathbb{R}^{n\times r} \text{ orthonormal,}$$

and $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_r)$. Consider a perturbation

$$\widehat{M} \;=\; M + E, \qquad \widehat{U} = [\hat{u}_{r+1}, \dots, \hat{u}_n], \quad \widehat{\Lambda} = \operatorname{diag}(\hat{\lambda}_{r+1}, \dots, \hat{\lambda}_n).$$

1.2 Eigengap hypothesis

Assume the target r-eigenvalue cluster lies in $[\alpha, \beta]$ while the rest of the spectrum lies outside a Δ-neighborhood:

$$\{\lambda_i\}_{i=1}^{r} \subset [\alpha, \beta], \qquad \{\lambda_i\}_{i=r+1}^{n} \subset (-\infty, \alpha - \Delta] \;\cup\; [\beta + \Delta, \infty)$$

for some $\Delta > 0$. (No global ordering is needed.)

1.3 Davis–Kahan $\sin\Theta$ theorem

$$\|\sin\Theta(\widehat{U}, U)\| \;\le\; \frac{\|E\|}{\Delta}$$

where $\Theta(\widehat{U}, U)$ is the principal angle matrix between the column spaces of $\widehat{U}$ and U, and $\|\cdot\|$ is the operator norm.

1.4 Proof sketch

Assume for simplicity (WLOG) that after centering the spectra by a scalar shift we have a symmetric window. Shift both matrices by the same constant:

$$\big(M - \tfrac{\alpha+\beta}{2} I,\; \widehat{M} - \tfrac{\alpha+\beta}{2} I\big) \qquad \text{(eigenvectors stay the same; eigenvalues shift).}$$

Let $U \in \mathbb{R}^{d\times r}$ span the target invariant subspace of M with $MU = U\Lambda$ (diagonal Λ), and let $\widehat{U} \in \mathbb{R}^{d\times r}$ span the corresponding subspace of $\widehat{M}$; write $\widehat{U}_\perp$ for an orthonormal basis of $\widehat{\mathcal{U}}^\perp$ and $\widehat{\Lambda}_\perp$ for the diagonal block of $\widehat{M}$ on $\widehat{\mathcal{U}}^\perp$:

$$\widehat{M}\widehat{U}_\perp = \widehat{U}_\perp\, \widehat{\Lambda}_\perp.$$

With the centered window $[-\beta, \beta]$ and gap $\Delta > 0$ separating it from the rest of the spectrum,

$$\|\Lambda\| \le \beta, \qquad \sigma_{\min}(\widehat{\Lambda}_\perp) \ge \beta + \Delta.$$

Let $E := \widehat{M} - M$. Project the invariance relations onto $\widehat{\mathcal{U}}^{\perp}$:

$$\begin{aligned} \widehat{U}_{\perp}^{\top}(\widehat{M} - M)U &= \widehat{U}_{\perp}^{\top}\widehat{M}U - \widehat{U}_{\perp}^{\top}MU \\ &= \widehat{\Lambda}_{\perp}\,\widehat{U}_{\perp}^{\top}U \;-\; \widehat{U}_{\perp}^{\top}U\,\Lambda. \end{aligned} \tag{$*$}$$

Using $\|AB\| \geq \sigma_{\min}(A)\|B\|$ and $\|B\Lambda\| \leq \|\Lambda\|\,\|B\|$,

$$\begin{aligned} \|\widehat{U}_{\perp}^{\top}EU\| &= \|\widehat{\Lambda}_{\perp}\,\widehat{U}_{\perp}^{\top}U - \widehat{U}_{\perp}^{\top}U\,\Lambda\| \\ &\geq \sigma_{\min}(\widehat{\Lambda}_{\perp})\,\|\widehat{U}_{\perp}^{\top}U\| - \|\Lambda\|\,\|\widehat{U}_{\perp}^{\top}U\| \\ &\geq (\beta + \Delta - \beta)\,\|\widehat{U}_{\perp}^{\top}U\| = \Delta\,\|\widehat{U}_{\perp}^{\top}U\| = \Delta\|\sin\Theta(\widehat{U}, U)\|. \end{aligned}$$

Hence,

$$\|\sin\Theta(\widehat{U}, U)\| \leq \frac{\|\widehat{U}_{\perp}^{\top}EU\|}{\Delta} \leq \frac{\|EU\|}{\Delta}$$

1.5 Alternative proof (resolvent / contour integral)

Another approach uses analytic functional calculus. Let Γ be a contour in the complex plane enclosing the eigenvalues $\{\lambda_1, \ldots, \lambda_r\}$ and excluding the rest.

$$UU^{\top} = \frac{1}{2\pi i}\oint_{\Gamma}(M - zI)^{-1}\,dz, \qquad \widehat{U}\widehat{U}^{\top} = \frac{1}{2\pi i}\oint_{\Gamma}(\widehat{M} - zI)^{-1}\,dz.$$

Subtracting and applying the resolvent identity

$$(\widehat{M} - zI)^{-1} - (M - zI)^{-1} = (\widehat{M} - zI)^{-1}\,E\,(M - zI)^{-1},$$

gives the Davis–Kahan inequality after bounding the contour integral. (This is the approach of Bhatia–Kato.)

1.6 Special case: ordered eigenvalues

Assume now $\lambda_1 \geq \cdots \geq \lambda_n$. By Weyl's inequality,

$$|\hat{\lambda}_i - \lambda_i| \leq \|E\| \quad \forall i.$$

If M is rank r so that $\lambda_{r+1} = \cdots = \lambda_n = 0$, then

$$\hat{\lambda}_{r+1} \leq \|E\|, \qquad \lambda_r \geq \hat{\lambda}_r - \|E\|.$$

Hence, if $\lambda_r > 4\|E\|$, then

$$\Delta \;\geq\; \frac{\lambda_r}{2}.$$

Consequently,

$$\|\sin\Theta(\widehat{U}, U)\| \;\lesssim\; \frac{\|E\|}{\lambda_r}$$

and a sufficient condition for consistency is

$$\frac{\|E\|}{\lambda_r} \;\xrightarrow{p}\; 0.$$

1.7 Wedin's Theorem (SVD perturbation)

Let

$$M = U\Sigma V^\top, \qquad \widehat{M} = M + E.$$

Then Wedin's theorem provides a $\sin\Theta$ bound for left/right singular subspaces:

$$\|\sin\Theta(\widehat{U}, U)\|,\ \|\sin\Theta(\widehat{V}, V)\| \ \le\ \frac{2\|E\|}{\Delta - \|E\|}$$

provided $\Delta > 2\|E\|$, where Δ is the singular value separation.

Proof idea: Embed into symmetric block matrices:

$$\begin{bmatrix} 0 & M \\ M^\top & 0 \end{bmatrix}, \qquad \begin{bmatrix} 0 & \widehat{M} \\ \widehat{M}^\top & 0 \end{bmatrix},$$

then apply Davis–Kahan.

1.8 Example: Balanced 2-block SBM eigen-structure

Consider a balanced two-community SBM with $p = a/n,\ q = b/n$ and block matrix $B = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$. Then

$$\lambda_1 \propto a + b, \qquad \lambda_2 \propto a - b,$$

so the eigengap $\sim |a - b|$, and $\|\sin\Theta(\widehat{U}, U)\| \lesssim \|E\|/|a - b|$.

2 Matrix Concentration via Exponential Moments

2.1 scalar Hoeffding's inequality

Let $X_1, \ldots, X_n$ be independent with $X_i \in [a, b]$ a.s. and let $\bar{X} = \frac{1}{n}\sum_i X_i$. Then, for $t > 0$,

$$\Pr\left(|\bar{X} - \mathbb{E}\bar{X}| \ge t\right) \ \le\ 2\exp\Big(-\frac{2n^2t^2}{\sum_{i=1}^n (b-a)^2}\Big) \ =\ 2\exp\Big(-\frac{2nt^2}{(b-a)^2}\Big).$$

2.1.1 Proof sketch

Chernoff bound (scalar). For any real $\theta > 0$ and random variable X,

$$\Pr[X > t] = \Pr\big[e^{\theta X} > e^{\theta t}\big] \ \le\ \frac{\mathbb{E}e^{\theta X}}{e^{\theta t}} \qquad \text{(Markov/Chernoff).}$$

If $X = \sum_{i=1}^n X_i$ with independent summands, then

$$\mathbb{E}e^{\theta X} = \mathbb{E}\prod_{i=1}^n e^{\theta X_i} = \prod_{i=1}^n \mathbb{E}e^{\theta X_i}.$$

(Optimize in θ.)

Hoeffding's lemma (centered bounded r.v.). If $X \in [a,b]$ a.s. and $\mathbb{E}X = 0$, then for all $s \in \mathbb{R}$,

$$\mathbb{E}\, e^{sX} \;\le\; \exp\Big(\tfrac{s^2}{8}\,(b-a)^2\Big).$$

Matrix exponential. For a (square) matrix M,

$$e^M \;=\; \sum_{i=0}^{\infty} \frac{M^{\,i}}{i!}, \qquad \text{(compare with scalar)}\;\; e^x = \sum_{i=0}^{\infty} \frac{x^i}{i!} = 1 + \sum_{i=1}^{\infty} \frac{x^i}{i!}.$$

Submultiplicativity and the series give the norm bound

$$\|e^M\| \;\le\; \sum_{i=0}^{\infty} \frac{\|M\|^i}{i!} \;=\; e^{\|M\|}.$$

A naive factorization fails:

$$\mathbb{E}\, e^{\theta \sum_i X_i} \;\neq\; \prod_i \mathbb{E}\, e^{\theta X_i},$$

because $(A+B)^2 = A^2 + AB + BA + B^2$ and $AB \neq BA$ in general. This is the main obstacle in lifting scalar mgf arguments to matrices.

Spectral calculus for symmetric matrices. If M is symmetric with spectral decomposition $M = U\Lambda U^\top$ (U orthonormal, $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_d)$), then

$$e^M = U\, e^{\Lambda} U^\top, \qquad e^{\Lambda} = \operatorname{diag}\big(e^{\lambda_1}, \dots, e^{\lambda_d}\big).$$

Tail bound for the top eigenvalue. Let X be (symmetric) random matrix and $\lambda_{\max}(X)$ its largest eigenvalue. For any $\theta > 0$,

$$\Pr\big[\lambda_{\max}(X) > t\big] = \Pr\big[\lambda_{\max}(\theta X) > \theta t\big] \;\le\; e^{-\theta t}\, \mathbb{E}\, e^{\lambda_{\max}(\theta X)} \;\le\; e^{-\theta t}\, \mathbb{E}\, \operatorname{tr}\Big(e^{\theta X}\Big).$$

2.2 Lieb's inequality

Theorem 1 (Lieb, 1973) *If H is self-adjoint and $A \succ 0$, then the map*

$$A \;\mapsto\; \operatorname{Tr}\, \exp\big(H + \log A\big)$$

is concave on the positive-definite cone. (Here $\log A$ is the matrix logarithm, well-defined for $A \succ 0$.)

Corollary 2 *For any random self-adjoint X,*

$$\mathbb{E}\, \operatorname{Tr}\big[\exp(H + X)\big] \;\le\; \operatorname{Tr}\, \exp\Big(H + \log \mathbb{E}\, e^X\Big).$$